

WEEK:- 9

Topic: Python Programming, File Handling, JSON Processing, and Exception Handling

Objectives:

To understand Python programming fundamentals, work with file handling operations, process JSON data, implement exception handling techniques, and develop efficient and reliable software solutions.

Learning Outcomes:

- Learned Python programming concepts, syntax, and control structures.
- Understood file handling operations for reading, writing, and managing data.
- Gained knowledge of JSON processing and data exchange mechanisms.
- Developed skills in exception handling and error management.
- Improved logical thinking and problem-solving abilities through practical programming tasks.

WORK SUMMARY:

During Week 9, I worked on Python programming, file handling, JSON processing, and exception handling. The training focused on understanding the core concepts of Python and their practical applications in software development. I attended technical sessions and participated in hands-on exercises to learn how Python can be used to develop efficient, scalable, and maintainable applications. Various programming examples and real-world scenarios were explored to strengthen my understanding of coding principles and development practices.

Throughout the week, I gained practical experience in writing Python programs using variables, data types, loops, conditional statements, functions, and modules. I learned how to perform file handling operations for storing and retrieving information from text files and other data sources. Special emphasis was placed on JSON processing, where I worked with structured data formats commonly used for application integration and data exchange. I also explored exception handling techniques to identify, manage, and resolve runtime errors effectively, ensuring application reliability and stability.

The activities carried out during this week enhanced my programming skills and strengthened my understanding of backend development concepts. Regular coding exercises, debugging sessions, and mentor feedback helped me improve my analytical thinking and troubleshooting abilities. I also learned the importance of writing clean, efficient, and well-documented code following industry best practices.

Activities Performed:

- Developed Python programs involving file handling, JSON processing, and data management.
- Implemented exception handling techniques and performed debugging of Python applications.